

Project#2

Functional Decomposition

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1. Own-choice quantity

- I have chosen Wolf who eats deer. Wolf eats 2 deers per month so needs twice the number of deer of its own number to increase its own species.

2. Table

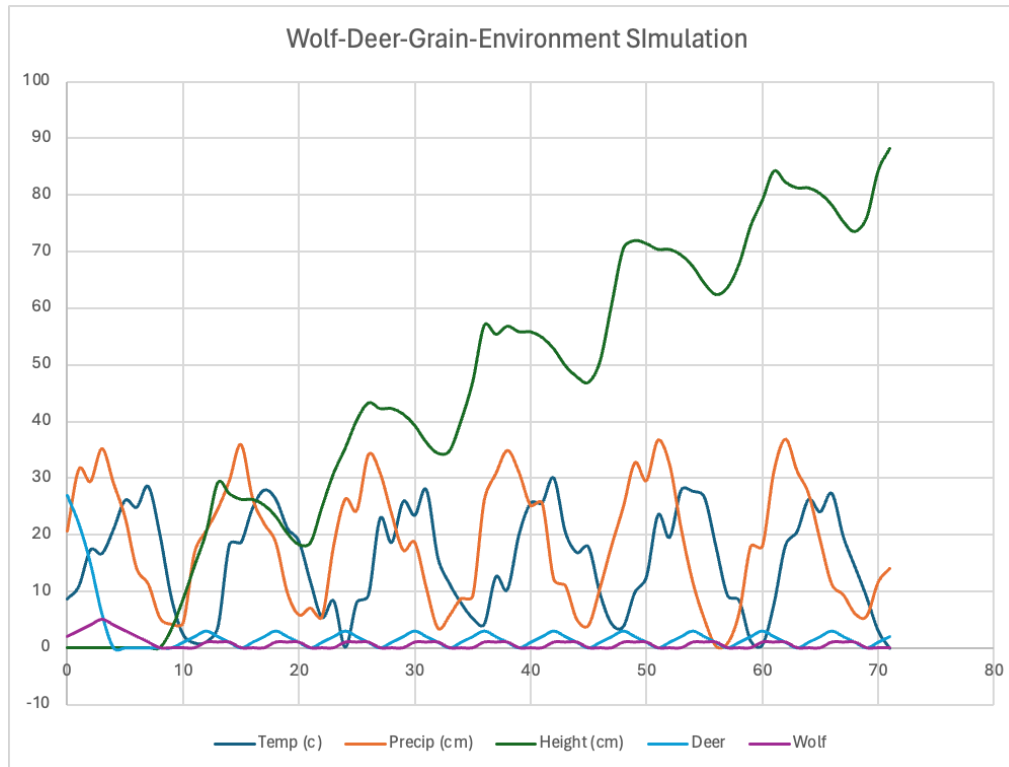
Deer-Grain-Environment Simulation

Months	Temp (c)	Precip (cm)	Height (cm)	Deer
0	8.6	20.65	0	29
1	10.84	31.59	0	28
2	17.25	29.43	0	27
3	16.6	35.23	0	26
4	20.94	29.1	0	25
5	26.04	23.03	0	24
6	24.79	13.97	0	23
7	28.44	11.23	0	22
8	19.94	5.27	0	21
9	8.7	4.15	0	20
10	2.32	4.39	0	19
11	0.79	16.93	0	18
12	1.01	20.72	0	17
13	3.59	24.58	0	16
14	18.22	29.64	0	15
15	18.58	35.95	0	14
16	24.66	26.48	0	13
17	27.82	21.97	0	12
18	26.22	18.64	0	11
19	21.11	9.76	0	10
20	18.73	5.8	0	9
21	11.57	7.04	0	8
22	5.29	5.5	0	7
23	8.24	18.09	0	6
24	0.04	26.29	0.4	5
25	7.99	24.35	3.37	4
26	9.26	34.16	4.4	3
27	22.77	30.97	1.4	4
28	18.57	23.68	0	3
29	25.82	17.29	0	2
30	23.38	18.62	0	1
31	27.88	10.57	0	0
32	15.84	3.46	0.08	0
33	11.29	5.7	1.52	0
34	7.84	8.71	6.89	1
35	5.17	9.15	13.72	2
36	4.13	26.11	23.68	3
37	12.49	30.8	22.09	4
38	10.28	34.92	21.54	5
39	19.99	31.02	16.55	6
40	25.5	25.16	10.55	7
41	25.62	25.59	3.55	8
42	29.95	12.11	0	7
43	20.4	11	0	6
44	16.77	4.96	0	5
45	17.75	3.96	0	4
46	9.45	10.65	0	3
47	4.15	18.14	8.03	2
48	3.69	24.92	17.81	3
49	9.75	32.73	19.25	4
50	12.49	29.61	16.69	5
51	23.43	36.77	11.69	6
52	19.5	32.43	5.69	7
53	27.86	21.03	0	6
54	27.62	11.61	0	5
55	26.59	5.05	0	4
56	17.84	0.28	0	3
57	9.21	0.79	0	2
58	8.33	6.16	2.15	1
59	1.15	17.97	8.9	2
60	0.41	17.93	13.4	3
61	7.65	31.2	18.57	4
62	18.07	36.92	14.59	5
63	20.48	31.31	9.6	6
64	26.19	27.22	3.6	7
65	24	19	0	6
66	27.23	11.24	0	5
67	19.55	9.36	0	4
68	14.27	6.04	0	3
69	8.95	5.55	0.38	2
70	2.97	11.58	6.7	1
71	-0.15	14.05	10.66	2

Wolf-Deer-Grain-Environment Simulation

Months	Temp (c)	Precip (cm)	Height (cm)	Deer	Wolf
0	8.6	20.65	0	27	2
1	10.84	31.59	0	22	3
2	17.25	29.43	0	15	4
3	16.6	35.23	0	6	5
4	20.94	29.1	0	0	4
5	26.04	23.03	0	0	3
6	24.79	13.97	0	0	2
7	28.44	11.23	0	0	1
8	19.94	5.27	0	0	0
9	8.7	4.15	3.32	0	0
10	2.32	4.39	8.55	1	0
11	0.79	16.93	14.52	2	0
12	1.01	20.72	20.43	3	1
13	3.59	24.58	29.14	2	1
14	18.22	29.64	27.16	1	1
15	18.58	35.95	26.18	0	0
16	24.66	26.48	26.18	1	0
17	27.82	21.97	25.18	2	0
18	26.22	18.64	23.18	3	1
19	21.11	9.76	20.18	2	1
20	18.73	5.8	18.19	1	1
21	11.57	7.04	18.56	0	0
22	5.29	5.5	24.91	1	0
23	8.24	18.09	30.83	2	0
24	0.04	26.29	35.23	3	1
25	7.99	24.35	40.2	2	1
26	9.26	34.16	43.23	1	1
27	22.77	30.97	42.23	0	0
28	18.57	23.68	42.25	1	0
29	25.82	17.29	41.25	2	0
30	23.38	18.62	39.25	3	1
31	27.88	10.57	36.25	2	1
32	15.84	3.46	34.33	1	1
33	11.29	5.7	34.77	0	0
34	7.84	8.71	40.14	1	0
35	5.17	9.15	46.97	2	0
36	4.13	26.11	56.93	3	1
37	12.49	30.8	55.34	2	1
38	10.28	34.92	56.79	1	1
39	19.99	31.02	55.8	0	0
40	25.5	25.16	55.8	1	0
41	25.62	25.59	54.8	2	0
42	29.95	12.11	52.8	3	1
43	20.4	11	49.8	2	1
44	16.77	4.96	47.84	1	1
45	17.75	3.96	46.86	0	0
46	9.45	10.65	50.67	1	0
47	4.15	18.14	60.7	2	0
48	3.69	24.92	70.48	3	1
49	9.75	32.73	71.92	2	1
50	12.49	29.61	71.36	1	1
51	23.43	36.77	70.36	0	0
52	19.5	32.43	70.36	1	0
53	27.86	21.03	69.36	2	0
54	27.62	11.61	67.36	3	1
55	26.59	5.05	64.36	2	1
56	17.84	0.28	62.38	1	1
57	9.21	0.79	63.63	0	0
58	8.33	6.16	67.78	1	0
59	1.15	17.97	74.53	2	0
60	0.41	17.93	79.03	3	1
61	7.65	31.2	84.2	2	1
62	18.07	36.92	82.22	1	1
63	20.48	31.31	81.22	0	0
64	26.19	27.22	81.22	1	0
65	24	19	80.22	2	0
66	27.23	11.24	78.22	3	1
67	19.55	9.36	75.23	2	1
68	14.27	6.04	73.52	1	1
69	8.95	5.55	75.9	0	0
70	2.97	11.58	84.22	1	0
71	-0.15	14.05	88.18	2	0

3. Graph



4. Analysis of the Graph.

- Settings

I initialized the height of grain as 5 inches (12.7 cm), the number of deer as 30, and the number of wolves as 1. The growth of grain is influenced by sufficient rain and heat, and its height decreases when deer consume it. The number of deer increases when there is enough grain to eat and decreases when they are hunted by wolves. Wolves, on the other hand, are affected only by the number of deer: when there are enough deer, the wolf population increases; when there are not enough deer to eat, the wolf population decreases.

- Analysis

- Grain Growth: The growth of grain is influenced by temperature and rainfall. These two factors follow cyclic patterns represented by cosine and sine functions. Consequently, the grain growth also follows this cyclic pattern, increasing and decreasing accordingly.

- Deer Population: The number of deer rapidly decreases because initially there is a large number of deer, and they quickly consume the available grain. Furthermore, with wolves preying on them, their population decreases rapidly.
- Wolf Population: The number of wolves remains low. As the number of deer decreases rapidly, there is not enough food for the wolves to sustain their population.
- Comparison with No-Wolf Simulation: Comparing this simulation with one where wolves are not included, it is evident that the presence of wolves significantly affects the simulation. The wolf population reduces the deer population, allowing the grain to grow more rapidly.